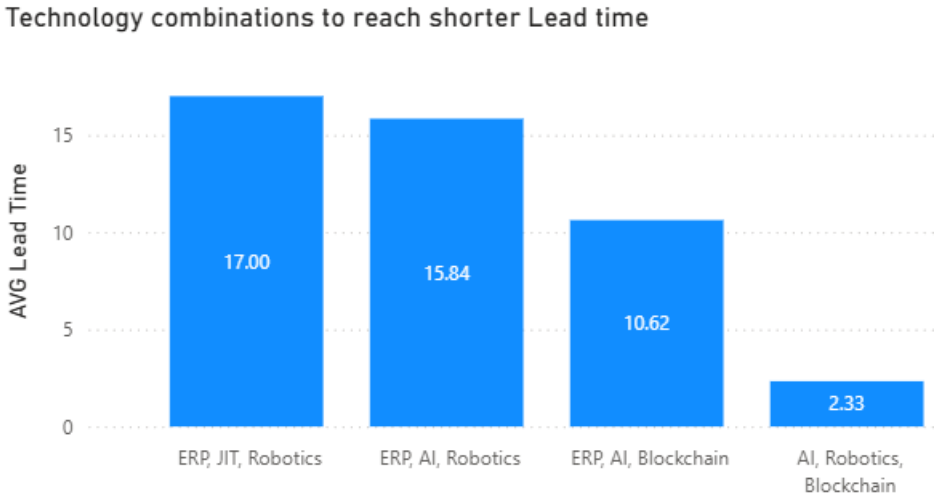
Data Visualization Pitch

Student Individual Assignment



Best Supply Chain Methodologies and Technology Approaches

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Research questions

- Related to company goals and industry; Which SCM practices and technology combinations most effectively reduce lead times and maximize customer satisfaction?

- How do SCM practices and technologies vary in their impact across different industrial sectors?

About Data

Supply chain practices according to different companies: *data link*

- *Acquiring comprehensive Supply Chain datasets is challenging, and some values may appear unrealistic.*
- *In this project, the supply chain practices and attributes of* **774 different companies** *are analyzed. For each company,* **23 different attributes** *are available;since supply chain is a complex field with many relationships, the analysis is made more realistic by having this many attributes.*
- *Including more companies was desired, but as mentioned, it is really hard to find such data.*
- *To make the findings more meaningful, the data was enriched with* **industry** *and* **country-level information**, *because supply chain processes are heavily impacted by industry.*
- *The data was received from* **Kaggle**, *and although the license status is unknown, it is being used strictly for educational purposes.*
- *The correctness of the data might be considered suspect because unrealistic columns are detected and outlined in this Project.*
- **Python** *was used for the data cleaning part and* **Power BI** *was used for analyzing and visualization.*

Methodology

- *After retriveving the data from Kaggle;*
- *Supplier count, Sustainability_Practices, Environmental_Impact_Score columns are deleted because the values are unrealistic and have no contribution to differentiation. COGS_Billion datatype is fixed with the Pandas library in Python.*
- *The values of Technology_utilized column were re-ordered and standardized to prevent misunderstandings.*
- *ChatGPT was used for code generation and for identifying the industries and countries of the companies. Descriptive statistics were applied, and correlation metrics were defined to determine which attributes have a critical impact. Since the lead time column is particularly important, the data were aggregated by industry to examine how the average lead time changes.*
- *AI-generated industry data was cross-referenced with company websites and LinkedIn profiles for 10% of the sample to ensure accuracy.*
- *In the data, Supplier count had unrealistic values which were not included in the analysis to maintain data integrity.For the Revenue Growth Rate, values higher than 15 were converted to 15.*
- *We have 6 different sectors and 8 supply chain practices that companies have been using.*
- *Project Source Code (GitHub)*

Insights from the Data

- *Attributes are defined by parameters such as percentage (%), days, billions, and scores.*
- *The dataset contains 999 rows but represents* **774 unique companies**; *some companies are repeated with different supply chain practices, which is acceptable since the focus is on practices and technologies over time.*
- *562 of the companies are based in the United States.*
- ***Core Sectors (56%)*** *: Automotive (30%) & Steels-Metals (26%)*
- ***Secondary Sectors (30%):*** *Agri-Chemicals (16%) & Tech-Software (14%)*
- ***Specialized Sectors (15%):*** *Pharma-Healthtech (13%) & Retail Consumer (2%)*
- *SCM Practices utilized include* ***33.82% Lean Manufacturing, 32.37% Agile SCM, and 17.63% Vendor Managed Inventory.***
- *In the descriptive analysis, the mean inventory turnover ratio is 6, while* **the maximum value is 50 days.** *14 companies are identified as outliers for this attribute.*
- *When the correlation of the data is examined, Lead Time is shown to affect 8 different attributes, which is expected in supply chain processes. Furthermore, a correlation ratio of 0.65 is identified between Transportation Cost Efficiency and Revenue Growth Rate.* ***Lead time one of the key attributes to increase Revenue.***
- *In terms of technology utilized,* **ERP, AI, and Blockchain are preferred by 775 entries**, *while Robotics is preferred over Blockchain by 157 companies .*
- *Another important part of Supply chain is* **Customer satisfaction.** *It provides companies with the power to increase order quantities and makes them the first candidate for new projects. Additionally, it creates significant opportunities to strengthen their position within the sector.*

Technology Choice

In the real world, once a company structure is formed over several years, reshaping these tools is often impossible or prohibitively expensive. ***Strategic investment in the right technology*** *is the key to reaching shorter lead times.*

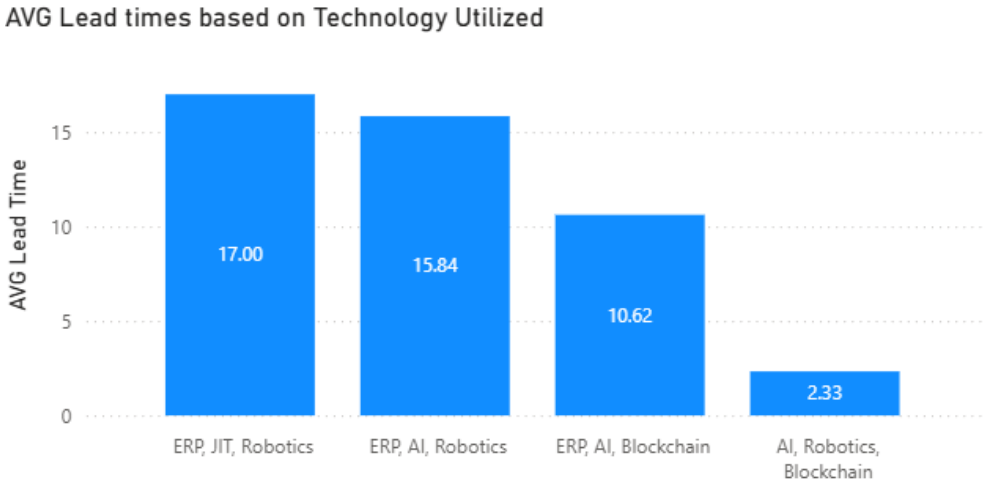
The ***overall mean lead time*** *is* ***11.45 days*** *HOWEVER; when we include the using technology combinations, Lead time changed significantly.*

- *A group of 32 companies utilizing a combination of* ***AI, Robotics, and Blockchain*** *achieved the lowest lead time in the dataset, averaging only* ***2.33 days****. Notably, these companies are spread across various industries.*

- *The most common configuration is observed among* ***613 different*** *companies across 6 industries using* ***ERP, AI, and Blockchain****. Using ERP instead of Robotics increases the lead time by* ***8 days.***

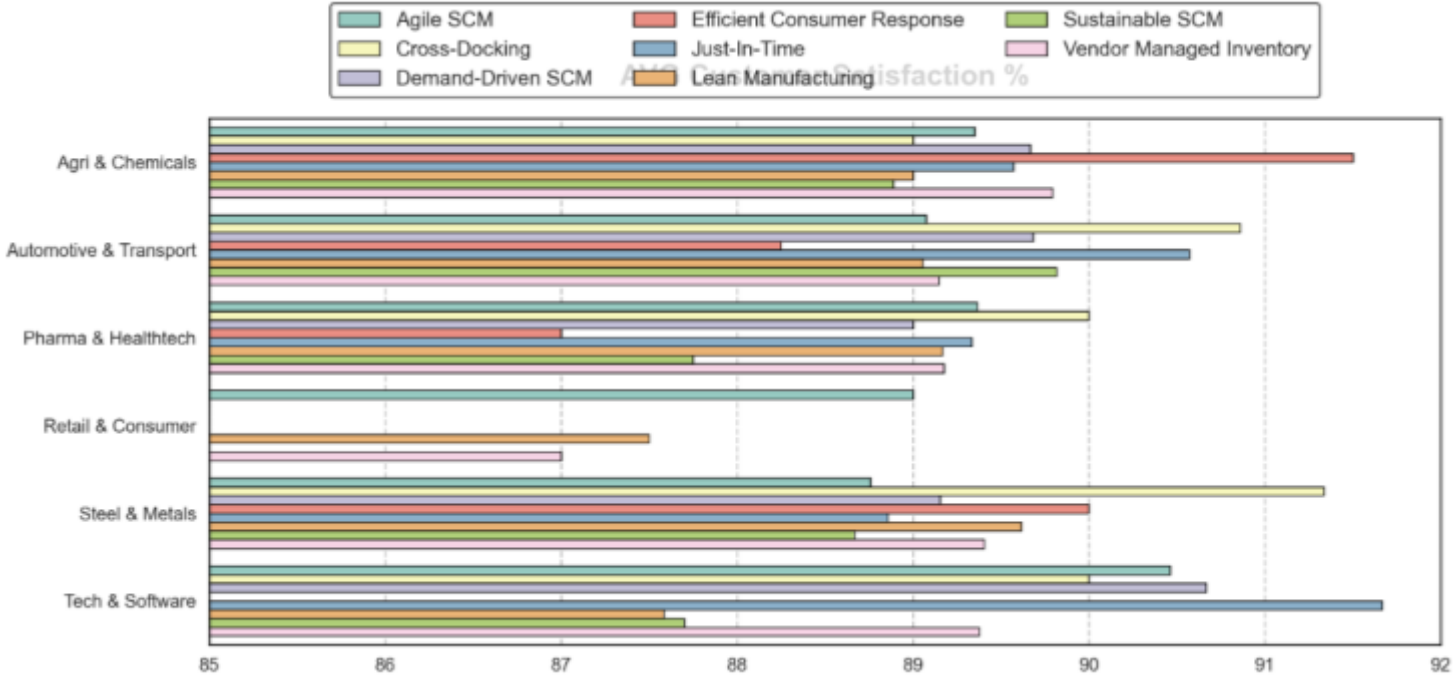
- ***144 companies utilizing ERP, AI, and Robotics*** *recorded a significantly higher lead time of* ***15.84 days.***

- *Finally,* ***43 companies relying on ERP, JIT (Just-In-Time)****, and* ***Robotics*** *were found to have a lead time of* ***17 days.***



Choosing Supply Chain Practice Based on Customer Satisfaction

- *In the Agri & Chemicals sector,* **Efficient Consumer Response (ECR)** *increases satisfaction to 91.50, which is more than 2 points above the general average.*



- *In the Tech & Software industry, the* **Just-in-Time (JIT)** *methodology has the highest impact, reaching a score of 91.67.*

- *For the Retail & Consumer industry, the available data is insufficient to make reliable predictions.*

- **Cross-Docking** *practice provides the highest customer satisfaction in Automotive & Transport (90.86),Pharma & Healthtech (90.00) and Steels & Metals (91.33)*

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